

Glucotwin

Predictive patient monitoring with
explainable 2-hour glucose risk forecasts

Random Forest | Simulated today, wearables next



PROJECT TITLE

Glucotwin



TEAM

BLACKBOX



MEMBER

Balaswamy Naik Devasothu



INSTITUTION

Rajiv Gandhi University of
Knowledge Technologies, Nuzvid



CONTACT

7670989763
balaswamy.devs@gmail.com

Problem Statement



Fragmented data

Patient monitoring data is spread across different systems.



Missed early warnings

Healthcare teams can miss early signs of a developing glucose rise.



Reactive follow-up

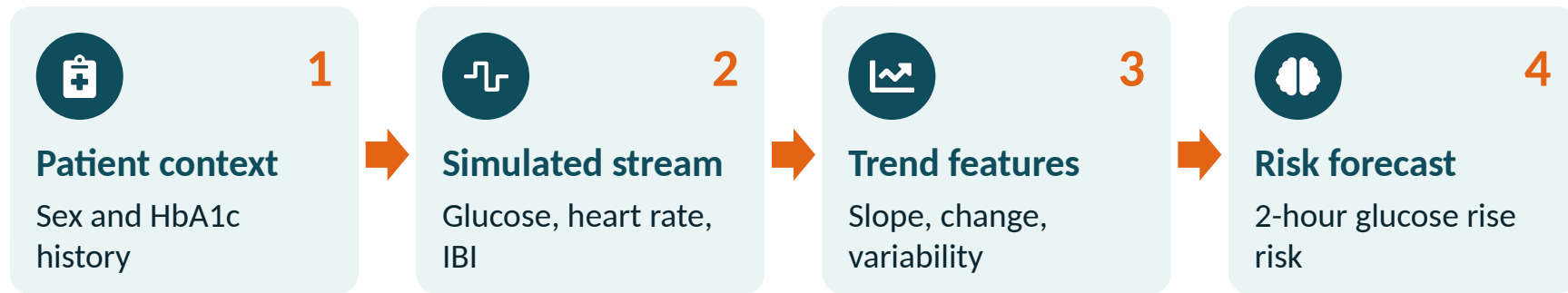
Without a forecast, it is hard to prioritize who to review first.

What clinicians need

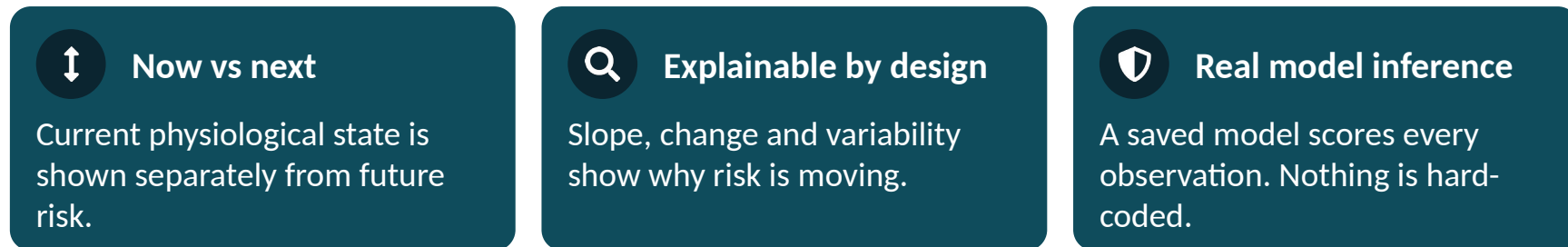
- ✓ Current physiological state
- ✓ Recent trends
- ✓ Historical context
- ✓ Explainable future risk

GlucoTwin brings all four into one interface.

Proposed Solution and Uniqueness



What makes it unique



Healthcare Use Case



Patient context

Sex and HbA1c shown alongside live data



Live observations

Simulated glucose, heart rate and IBI today



Trend windows

Glucose trends over 15, 30 and 60 minutes



Explained risk

Slope, change and variability show why risk moves



2-hour prediction

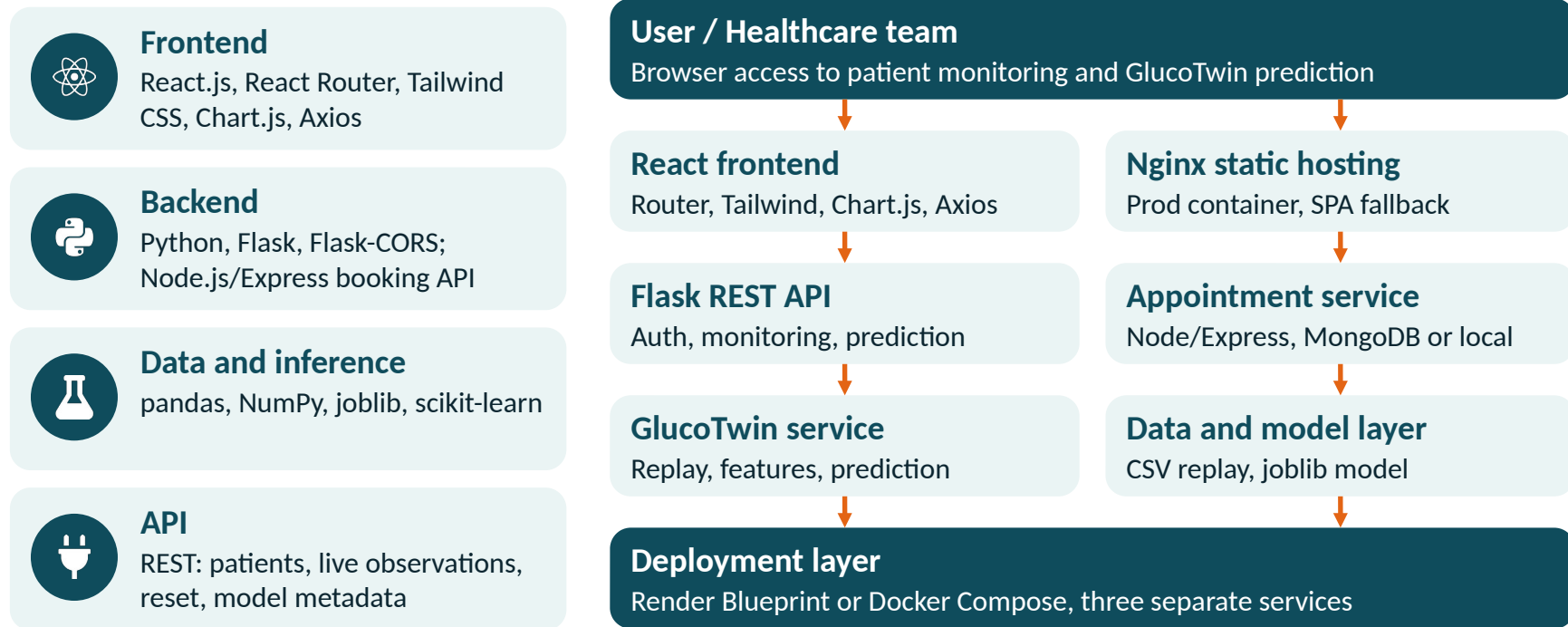
Risk of a glucose rise of at least 30 mg/dL



Prioritized review

Helps teams decide which patients to follow up first

Tech Stack and System Architecture



AI/ML Methodology and Workflow

- 1 **Historical profile:** patient ID, sex and HbA1c are loaded
- 2 **Simulated observation:** glucose, heart rate and IBI advance through the time series
- 3 **Temporal features:** 15/30/60-minute glucose change, slope and variability
- 4 **Model inference:** saved Random Forest pipeline gives predict_proba output
- 5 **Future 2-hour risk:** glucose rise of 30+ mg/dL within the next 2 hours
- 6 **Explainable UI:** state, trends, risk level and why risk is changing

PREDICTION TARGET

30+ mg/dL

glucose rise within the next 2 hours

The model is not retrained inside the app, and predictions are never hard-coded.

Application Modules



Login

Users sign in before opening patient views



Patient monitoring

Patient history with simulated glucose, heart rate and IBI



GlucoTwin prediction

2-hour glucose rise risk from the Random Forest



Trend and explainability

Slope, change and variability over 15/30/60 min



Appointments

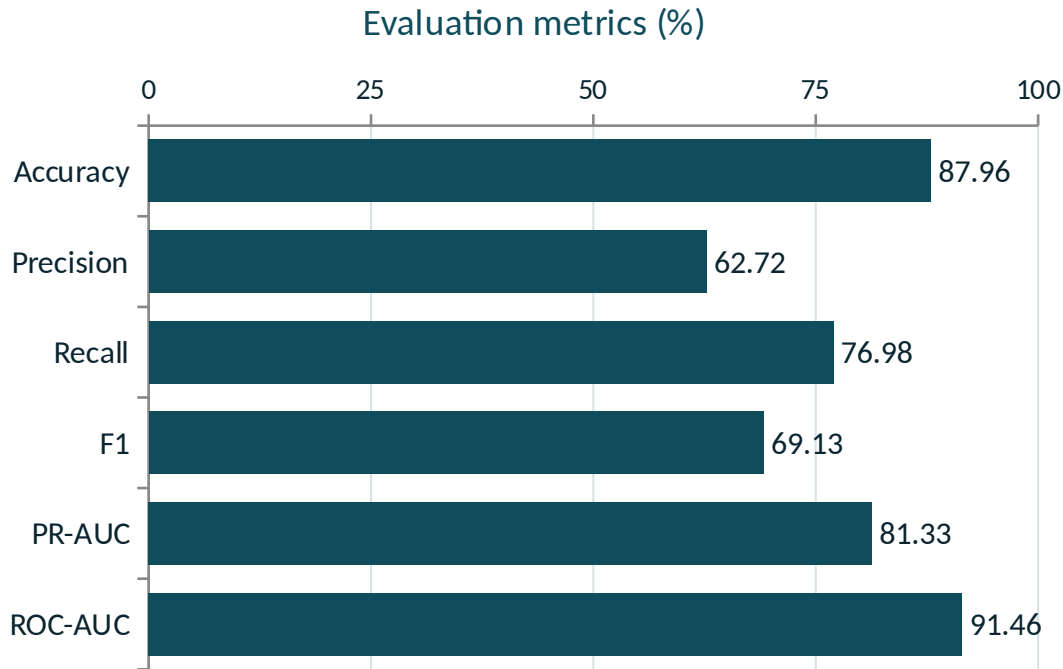
Node.js/Express booking API with MongoDB



AI chatbot

Assistant integrated through the Hugging Face API

Model Evaluation Metrics



91.46%

ROC-AUC: strong ranking of risky vs safe moments

87.96%

Accuracy across evaluated observations

76.98%

Recall: roughly 3 in 4 glucose rises flagged

Feasibility, Impact, Limits, Future Scope



Feasibility

- Lightweight stack: React, Flask, scikit-learn
- Dockerized with one-command compose
- Render Blueprint for cloud deploy



Impact

- One view of state, trends and risk
- Helps prioritize patients for follow-up
- Explainable signals build trust



Limitations

- Data is simulated replay, not live sensors
- Not clinically validated
- Precision is 62.72%, so false alerts occur



Future scope

- Real-time wearable sensors plus history
- Prospective clinical validation
- Broader models and personalization

Conclusion and References

Conclusion

- ✓ GlucoTwin unifies history, live trends and forecast in one view
- ✓ A Random Forest scores 2-hour glucose rise risk with 91.46% ROC-AUC
- ✓ Explainable signals make each risk change understandable
- ✓ A deployable prototype, ready for real-time wearable data

References

GlucoTwin repository

github.com/BalaswamyNaikD/GlucoTwin

scikit-learn

scikit-learn.org

Flask

flask.palletsprojects.com

React

react.dev

Hugging Face

huggingface.co